

Problems (Total: 30 Points)

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

Activating project at `c:\Users\Pranati\OneDrive - Cornell University\Documents\BEE4750-Gi

```
using JuMP
using HiGHS
using DataFrames
using Plots
using Measures
using CSV
using MarkdownTables
```

Problem 1 (Total: 3 Points)

Solve the following linear program **analytically**. Show your work and justify any reasoning.

$$\begin{aligned} \max_{x,y} \quad & 3x + 2y \\ \text{subject to:} \quad & x + 4y \leq 16 \\ & x - 2y \leq -1 \\ & 0 \leq x \leq 4 \\ & 1 \leq y \leq 4. \end{aligned}$$

Problem 2 (12 points)

A farmer has access to a pesticide which can be used on corn, soybeans, and wheat fields, and costs \$70/kg-yr to apply. The crop yields the farmer can obtain following crop yields by applying varying rates of pesticides to the field are shown in Table 1.

Application Rate (kg/ha)	Soybean (kg/ha)	Wheat (kg/ha)	Corn (kg/ha)
0	2900	3500	5900
1	3800	4100	6700
2	4400	4200	7900

Table 1: Crop yields from applying varying pesticide rates for Problem 1.

The costs of production, *excluding pesticides*, for each crop, and selling prices, are shown in Table 2.

Crop	Production Cost (\$/ha-yr)	Selling Price (\$/kg)
Soybeans	350	0.36
Wheat	280	0.27
Corn	390	0.22

Table 2: Costs of crop production, excluding pesticides, and selling prices for each crop.

Recently, environmental authorities have declared that farms cannot have an *average* application rate on soybeans, wheat, and corn which exceeds 0.8, 0.7, and 0.6 kg/ha, respectively. The farmer has asked you for advice on how they should plant crops and apply pesticides to maximize profits over 130 total ha while remaining in regulatory compliance if demand for each crop (which is the maximum the market would buy) this year is 250,000 kg?

Problem 2.1

Formulate a linear program for this resource allocation problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations). **Tip: Make sure that all of your constraints are linear.**

Problem 2.2

How many ha should the farmer dedicate to each crop and with what pesticide application rate(s)? How much profit will the farmer expect to make?

Problem 2.3

The farmer has an opportunity to buy an extra 10 ha of land. How much extra profit would this land be worth to the farmer? Discuss why this value makes sense and under what conditions you would recommend the farmer should make the purchase.

Problem 2 (15 points)

For this problem, we will use hourly load (demand) data from 2013 in New York's Zone C (which includes Ithaca). The load data is loaded and plotted below in Figure 1.

Problem 1

$$\begin{aligned} \max \quad & 3x + 2y \\ \text{subject to: } & x + 4y \leq 16 \quad \textcircled{A} \\ & x - 2y \leq -1 \quad \textcircled{B} \\ & 0 \leq x \leq 4 \quad \textcircled{C} \\ & 1 \leq y \leq 4 \quad \textcircled{D} \end{aligned}$$

Plot on Desmos:

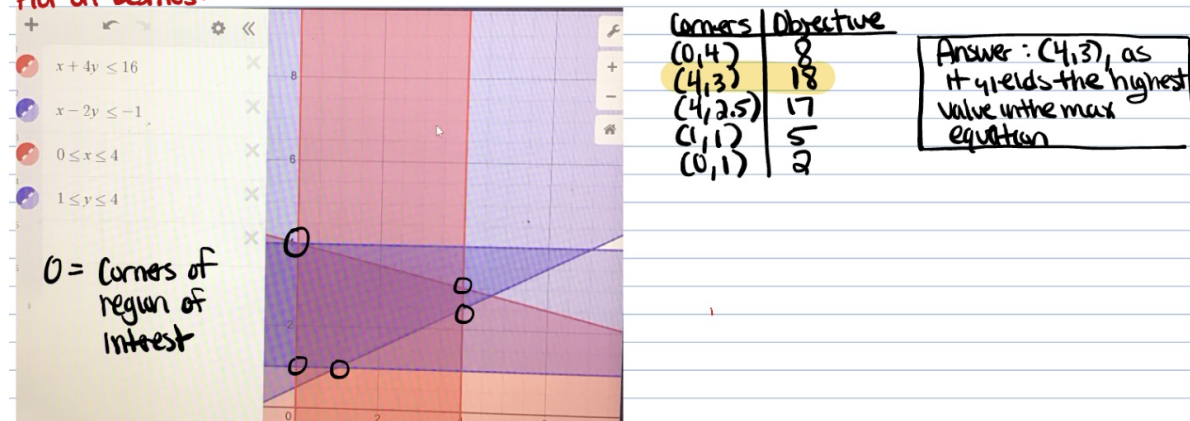
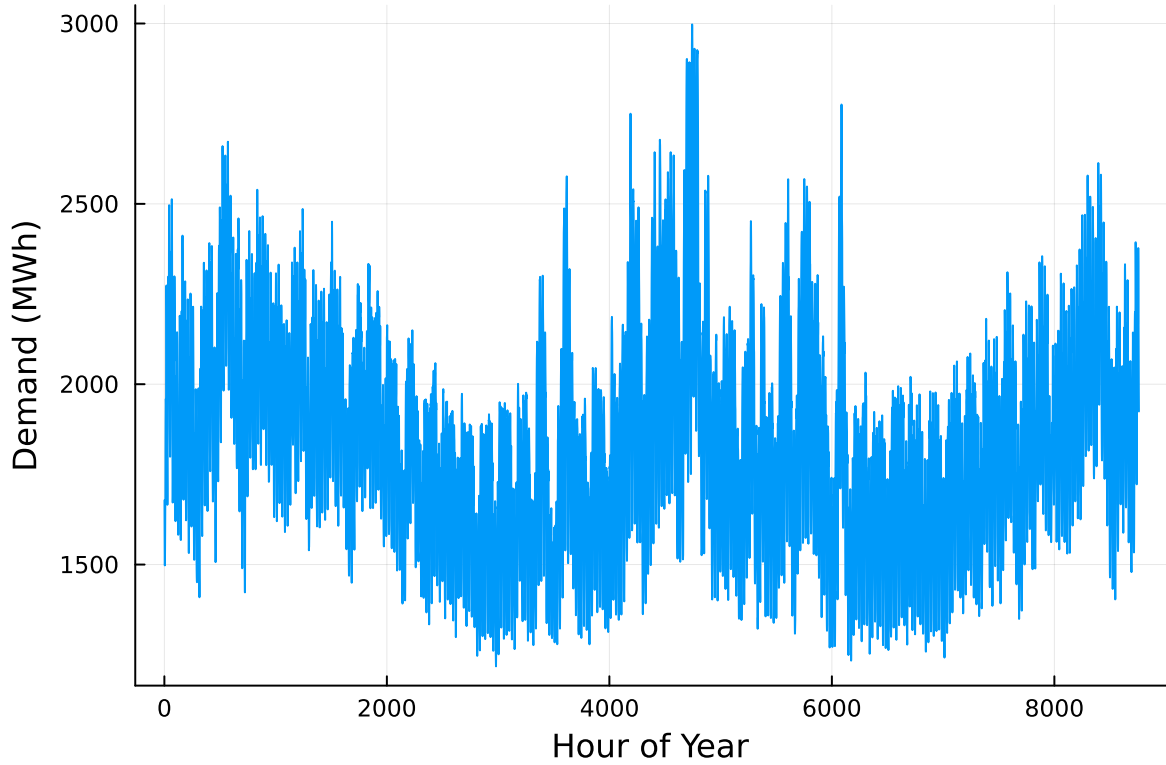


Figure 1: image.png

Problem 2

```
# load the data, pull Zone C, and reformat the DataFrame
NY_demand = DataFrame(CSV.File("data/2013_hourly_load_NY.csv"))
rename!(NY_demand, "Time Stamp" => :Date)
demand = NY_demand[:, [:Date, :C]]
rename!(demand, :C => :Demand)
demand[:, :Hour] = 1:nrow(demand)
```

```
# plot demand
plot(demand.Hour, demand.Demand, xlabel="Hour of Year", ylabel="Demand (MWh)", label=:false)
```



Next, we load the generator data, shown in Table 3. This data includes fixed costs (\$/MW installed), variable costs (\$/MWh generated), and CO2 emissions intensity (tCO2/MWh generated).

```
gens = DataFrame(CSV.File("data/generators.csv"))
```

	Plant	FixedCost	VarCost	Emissions
	String15	Int64	Int64	Float64
1	Geothermal	450000	0	0.0
2	Coal	220000	24	1.0
3	NG CCGT	82000	30	0.43
4	NG CT	65000	40	0.55
5	Wind	91000	0	0.0
6	Solar	70000	0	0.0

Finally, we load the hourly solar and wind capacity factors, which are plotted in Figure 2. These tell us the fraction of installed capacity which is expected to be available in a given hour for generation (typically based on the average meteorology).

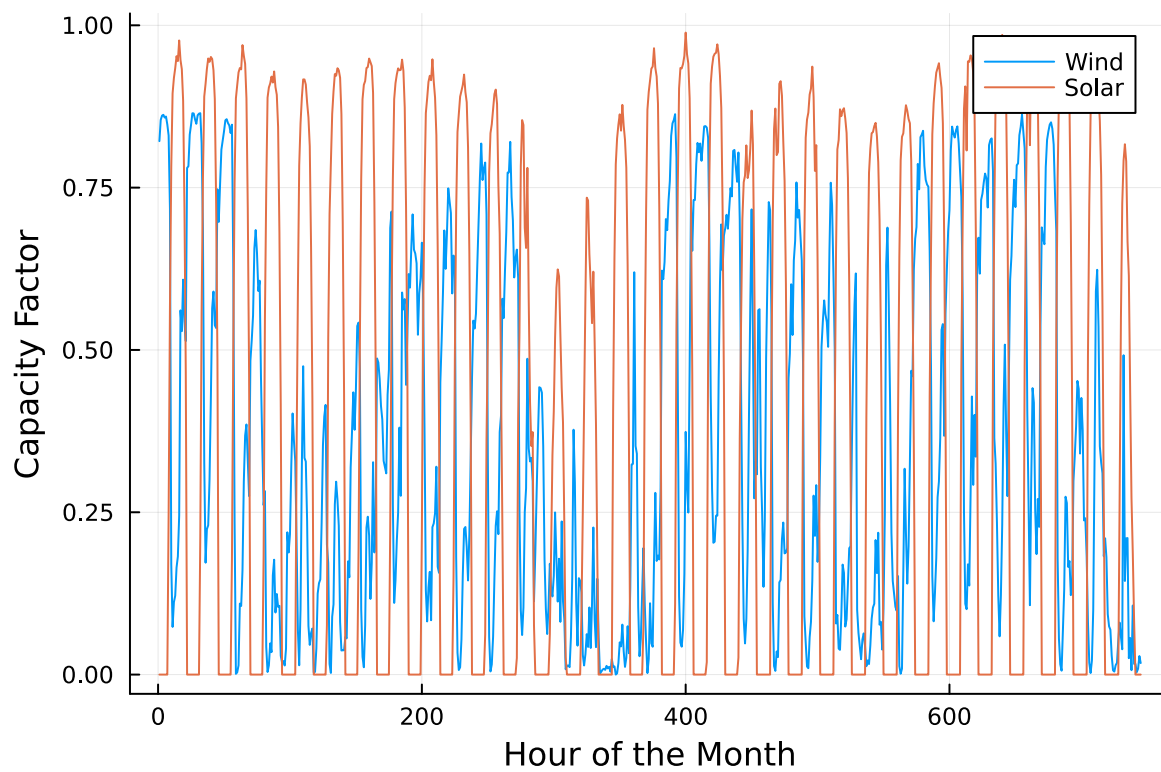
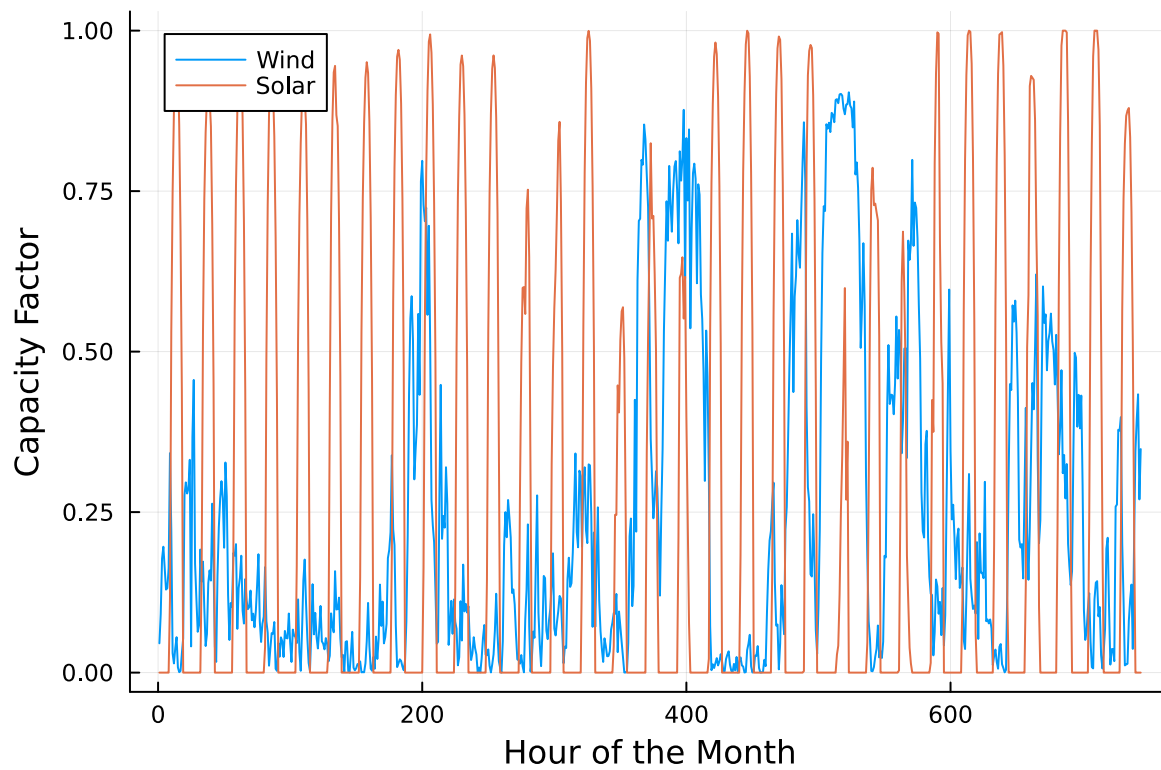
```
# load capacity factors into a DataFrame
cap_factor = DataFrame(CSV.File("data/wind_solar_capacity_factors.csv"))

# plot January capacity factors
p1 = plot(cap_factor.Wind[1:(24*31)], label="Wind")
plot!(cap_factor.Solar[1:(24*31)], label="Solar")
axis!("Hour of the Month")
axis!("Capacity Factor")

p2 = plot(cap_factor.Wind[4344:4344+(24*31)], label="Wind")
plot!(cap_factor.Solar[4344:4344+(24*31)], label="Solar")
axis!("Hour of the Month")
axis!("Capacity Factor")

display(p1)
display(p2)
```

Problem 2



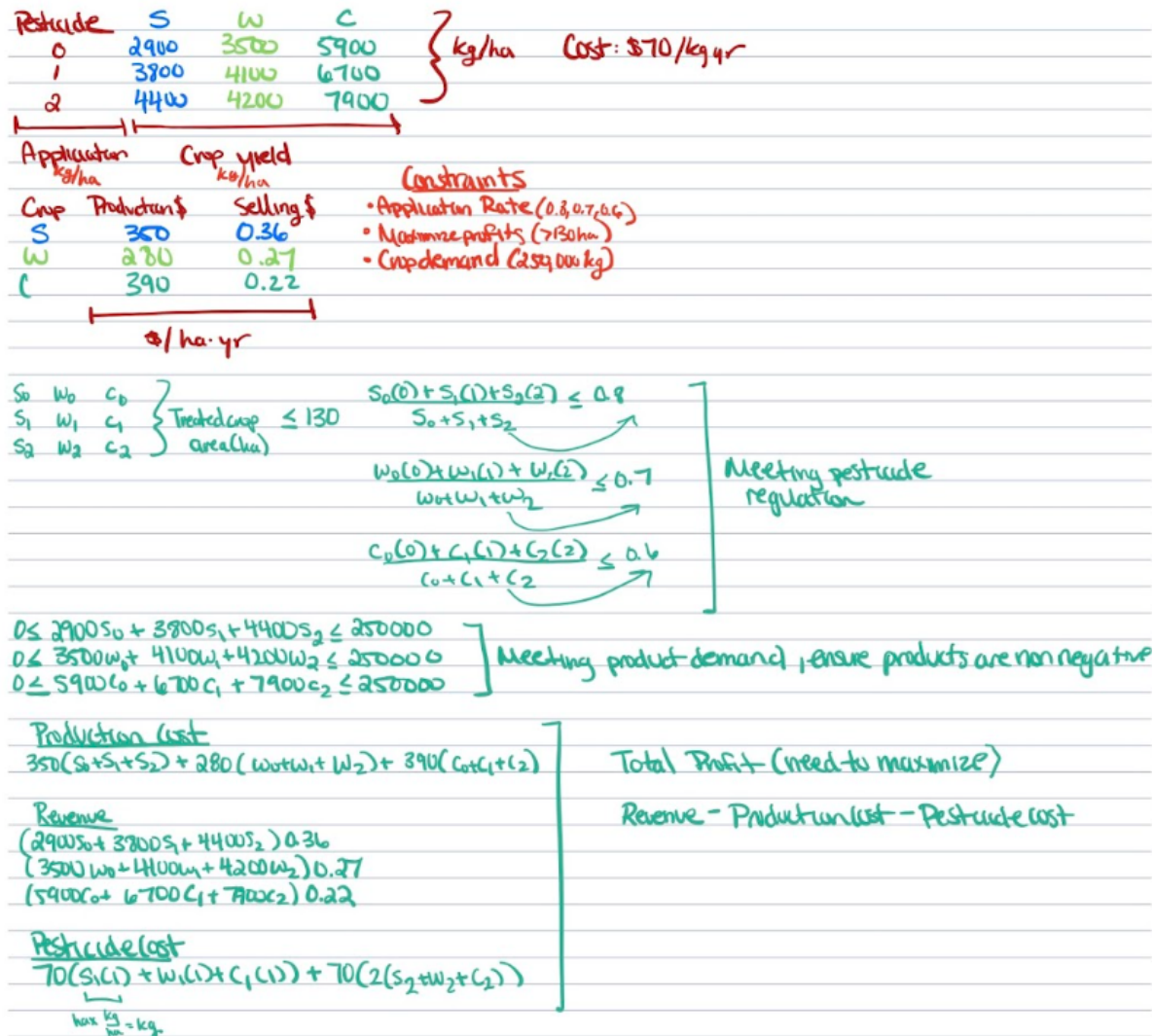


Figure 2: image.png

```
farm_model = Model(HiGHS.Optimizer)

#initialize each crop area as a variable
@variable(farm_model, crop[1:9] >= 0) #make sure crop area is not negative
s0 = crop[1]
s1 = crop[2]
s2 = crop[3]
w0 = crop[4]
w1 = crop[5]
```

```

w2 = crop[6]
c0 = crop[7]
c1 = crop[8]
c2 = crop[9]

s_sum = s0 + s1 + s2
w_sum = w0 + w1 + w2
c_sum = c0 + c1 + c2

#objective to maximize profit
ProdCost = 350*(s0 + s1 + s2) + 280*(w0 + w1 + w2) + 390*(c0 + c1 + c2)
Rev = 0.36*(2900*s0 + 3800*s1 + 4400*s2) + 0.27*(3500*w0 + 4100*w1 + 4200*w2) + 0.22(5900*c0
PestCost = 70*(s1 + w1 + c1 + 2*(s2 + w2 + c2)) #ignore s0, w0, and c0 since no pesticide is
@objective(farm_model, Max, Rev - ProdCost - PestCost)

#constraints
#meet crop area max + non negative
@constraint(farm_model, Crop_Area, (s_sum + w_sum + c_sum) <= 130)
#meet pesticide regulations
@constraint(farm_model, Soybean_Pesticide_Application, (s0*0 + s1*1 + s2*2) <= 0.8*s_sum)
@constraint(farm_model, Wheat_Pesticide_Application, (w0*0 + w1*1 + w2*2) <= 0.7*w_sum)
@constraint(farm_model, Corn_Pesticide_Application, (c0*0 + c1*1 + c2*2) <= 0.6*c_sum)
#meet product demand + non negative
@constraint(farm_model, Soybean_Demand, 0 <= 2900*s0 + 3800*s1 + 4400*s2 <= 250000)
@constraint(farm_model, Wheat_Demand, 0 <= 3500*w0 + 4100*w1 + 4200*w2 <= 250000)
@constraint(farm_model, Corn_Demand, 0 <= 5900*c0 + 6700*c1 + 7900*c2 <= 250000)

print(farm_model)
optimize!(farm_model)

```

Max 694 crop[1] + 948 crop[2] + 1094 crop[3] + 665.0000000000001 crop[4] + 757 crop[5] + 714

Subject to

Crop_Area : crop[1] + crop[2] + crop[3] + crop[4] + crop[5] + crop[6] + crop[7] + crop[8] +

Soybean_Pesticide_Application : -0.8 crop[1] + 0.19999999999999996 crop[2] + 1.2 crop[3] <=

Wheat_Pesticide_Application : -0.7 crop[4] + 0.30000000000000004 crop[5] + 1.3 crop[6] <= 0

Corn_Pesticide_Application : -0.6 crop[7] + 0.4 crop[8] + 1.4 crop[9] <= 0

Soybean_Demand : 2900 crop[1] + 3800 crop[2] + 4400 crop[3] in [0, 250000]

Wheat_Demand : 3500 crop[4] + 4100 crop[5] + 4200 crop[6] in [0, 250000]

Corn_Demand : 5900 crop[7] + 6700 crop[8] + 7900 crop[9] in [0, 250000]

crop[1] >= 0

crop[2] >= 0

crop[3] >= 0


```

crop[4] >= 0
crop[5] >= 0
crop[6] >= 0
crop[7] >= 0
crop[8] >= 0
crop[9] >= 0
Running HiGHS 1.11.0 (git hash: 364c83a51e): Copyright (c) 2025 HiGHS under MIT licence terms
LP has 7 rows; 9 cols; 27 nonzeros
Coefficient ranges:
  Matrix [2e-01, 8e+03]
  Cost   [7e+02, 1e+03]
  Bound  [0e+00, 0e+00]
  RHS    [1e+02, 2e+05]
Presolving model
7 rows, 9 cols, 27 nonzeros 0s
7 rows, 9 cols, 27 nonzeros 0s
Presolve : Reductions: rows 7(-0); columns 9(-0); elements 27(-0) - Not reduced
Problem not reduced by presolve: solving the LP
Using EKK dual simplex solver - serial
  Iteration      Objective      Infeasibilities num(sum)
          0      -1.8169844781e+02 Ph1: 7(28.3601); Du: 9(181.698) 0s
          7       1.1674116702e+05 Pr: 0(0) 0s
Model status      : Optimal
Simplex iterations: 7
Objective value    : 1.1674116702e+05
P-D objective error : 6.2325284535e-17
HiGHS run time     : 0.02

```

```

#2.2
#Dedicated Crop Area
@show value.(crop);

#Allocated pesticide application
@show shadow_price.(Soybean_Pesticide_Application);
@show shadow_price.(Wheat_Pesticide_Application);
@show shadow_price.(Corn_Pesticide_Application);

#Max profit
println("Max Profit is ", @show objective_value(farm_model))

```

```

value.(crop) = [13.812154696132593, 55.24861878453038, 0.0, 6.743306417339564, 15.7343816404,
shadow_price.(Soybean_Pesticide_Application) = 212.28176795580112

```

```
shadow_price.(Wheat_Pesticide_Application) = 91.99999999999989
shadow_price.(Corn_Pesticide_Application) = 108.67692307692312
objective_value(farm_model) = 116741.16702082448
Max Profit is 116741.16702082448
```

Problem 2.2 Write Up

According to the linea model, the allocated crop values are as follows

Soybean: $s_0 = 13.8$ ha $s_1 = 55.2$ ha $s_2 = 0$ ha

Wheat: $w_0 = 6.7$ ha $w_1 = 15.7$ ha $w_2 = 0$ ha

Corn: $c_0 = 26.9$ ha $c_1 = 0$ ha $c_2 = 11.5$ ha

In terms of total crop allocated pesticide applications, soybean was 212.3 kg/ha, wheat was 92.0 kg/ha, and corn was 108.7 kg/ha

In terms of total profit, the farmers get about \$116,741.17

```
#2.3
@show shadow_price(Crop_Area)
```

```
shadow_price(Crop_Area) = 729.4000000000001
```

```
729.4000000000001
```

Problem 2.3 Write Up

As seen from the model, the shadow price of the total crop areas is 729.4, representing the amount the solution (profit) would increase per unit (ha). Therefore, if we increase by 10 units of ha, the profit would increase by 729.4×10 dollars, or \$7,294. Rerunning the model with the increased ha in the constraint actually showed the profit to go up by this number.

You have been asked to develop a generating capacity expansion plan for the utility in Riley County, NY, which currently has no existing electrical generation infrastructure. The utility can build any of the following plant types: geothermal, coal, natural gas combined cycle gas turbine (CCGT), natural gas combustion turbine (CT), solar, and wind. The NY state legislature is planning to enact an annual CO₂ limit, which for the utility would limit the emissions in its footprint to 1.5 MtCO₂/yr.

While coal, CCGT, and CT plants can generate at their full installed capacity, geothermal plants operate at maximum 85% capacity, and solar and wind available capacities vary by the hour depend on the expected meteorology. The utility will also penalize any non-served demand at a rate of \$10,000/MWh.

Problem 3.1

Formulate a linear program for this capacity expansion problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations).

Problem 3.2

How much should the utility build of each type of generating plant? What will the total cost be? How much energy will be non-served?

Problem 3.3

What fraction of annual generation does each plant type produce? How does this compare to the breakdown of built capacity that you found in Problem 3.2? Do these results make sense given the demand and generator data?

Problem 3.4

Make a plot of the electricity price in each hour. Discuss any trends that you see.

Problem 3.5

What is the shadow price of CO₂? What is the value to the utility be of allowing it to emit an additional 1000 tCO₂/yr?

Significant Digits

Use `round(x; digits=n)` to report values to the appropriate precision! If your number is on a different order of magnitude and you want to round to a certain number of significant digits, you can use `round(x; sigdigits=n)`.

Getting Variable Output Values

`value.(x)` will report the values of a JuMP variable `x`, but it will return a special container which holds other information about `x` that is useful for JuMP. This means that you can't use this output directly for further calculations. To just extract the values, use `value.(x).data`.

Suppressing Model Command Output

The output of specifying model components (variable or constraints) can be quite large for this problem because of the number of time periods. If you end a cell with an `@variable` or `@constraint` command, I *highly* recommend suppressing

output by adding a semi-colon after the last command, or you might find that your notebook crashes.

3.1

decision vars:

→ C_p : installed capacity (MW)
per plant

→ $G_{p,t}$: generation (MWh)
by plant @ specific time

→ U_t : unserved demand in
hour t (MWh)

objective Function:

$$\min Z = \sum_p F_p C_p + \sum_t \sum_p V_p G_{p,t} + P_{\text{vol}} \sum_t U_t$$

Minimize total annual system cost

Constraints:

Hourly energy balance

$$\sum_p G_{p,t} + V_t = D_t$$

Capacity limits: all multiplied by CF(a_p)
coal, NG, CCGT, NGCT:

$$G_{p,t} \leq C_p a_{p,t}$$

geothermal:

$$G_{geo,t} \leq 0.85 C_g a_{geo}$$

wind/solar:

$$G_{ws,t} \leq a_{ws,t} C_{ws}$$

annual energy balance

$$\sum_t \sum_p G_{p,t} + V_t = D_t$$

nonnegativity

$$C_p \geq 0$$

F_p : fixed cost

V_p : variable cost

$a_{p,t}$: availability

E_p : emission

D_t : demand

P_{fuel} : fuel price

```

#3.2
#loading data -- the VSCode notebook helped with syntax

demand = DataFrame(CSV.File("data/2013_hourly_load_NY.csv"))
rename!(demand, :Time Stamp => :Date)
rename!(demand, :C => :Demand)
demand.Hour = 1:nrow(demand)
numHours = nrow(demand)

gens = DataFrame(CSV.File("data/generators.csv"))
CF = DataFrame(CSV.File("data/wind_solar_capacity_factors.csv"))
plants = gens.Plant

availability = Dict(
    "Geothermal" => fill(0.85, numHours),
    "Coal"       => fill(1.0, numHours),
    "NG CCGT"    => fill(1.0, numHours),
    "NG CT"      => fill(1.0, numHours),
    "Wind"       => CF.Wind,
    "Solar"      => CF.Solar
)

CO2cap = 1.5e6
penalty = 10000.0

fixed = Dict(gens.Plant .=> gens.FixedCost)
var = Dict(gens.Plant .=> gens.VarCost)
emi = Dict(gens.Plant .=> gens.Emissions)

#optimization model
model = Model(HiGHS.Optimizer)

@variable(model, C[p in plants] >= 0)
@variable(model, G[p in plants, t in 1:numHours] >= 0)
@variable(model, U[t in 1:numHours] >= 0)

@objective(model, Min,
    sum(fixed[p] * C[p] for p in plants) +
    sum(var[p] * G[p,t] for p in plants, t in 1:numHours) +
    penalty * sum(U[t] for t in 1:numHours)
)

```

```

#constraints
for t in 1:numHours
    @constraint(model,
        sum(G[p,t] for p in plants) + U[t] == demand.Demand[t]
    )
end

#capacity limit
for p in plants, t in 1:numHours
    @constraint(model,
        G[p,t] <= C[p] * availability[p][t])
end

#CO2 emissions constraint
@constraint(model, co2_limit, sum(emi[p] * G[p,t] for p in plants, t in 1:numHours) <= CO2cap)

#solve
optimize!(model)

```

Running HiGHS 1.11.0 (git hash: 364c83a51e): Copyright (c) 2025 HiGHS under MIT licence terms

LP has 61321 rows; 61326 cols; 188256 nonzeros

Coefficient ranges:

Matrix	[5e-05, 1e+00]
Cost	[2e+01, 4e+05]
Bound	[0e+00, 0e+00]
RHS	[1e+03, 2e+06]

Presolving model

56857 rows, 56862 cols, 179328 nonzeros 0s

Dependent equations search running on 8760 equations with time limit of 1000.00s

Dependent equations search removed 0 rows and 0 nonzeros in 0.00s (limit = 1000.00s)

56857 rows, 56862 cols, 179328 nonzeros 0s

Presolve : Reductions: rows 56857(-4464); columns 56862(-4464); elements 179328(-8928)

Solving the presolved LP

Using EKK dual simplex solver - serial

Iteration	Objective	Infeasibilities num(sum)
0	0.0000000000e+00	Pr: 8760(5.18855e+06) 0s
20724	4.5856896406e+08	Pr: 10167(5.2419e+06); Du: 0(1.88022e-07) 5s
26975	7.0089057297e+08	Pr: 9694(4.55027e+06); Du: 0(7.07016e-08) 11s
29777	7.7328394995e+08	Pr: 10324(3.24081e+06); Du: 0(7.07016e-08) 16s
32251	7.8521940971e+08	Pr: 650(14884.5); Du: 0(1.48437e-07) 22s
32698	7.8535295894e+08	Pr: 0(0) 22s

```

Solving the original LP from the solution after postsolve
Model status      : Optimal
Simplex iterations: 32698
Objective value    : 7.8535295894e+08
P-D objective error : 1.3357583179e-14
HiGHS run time     : 22.65

```

```

println("total system cost = ", objective_value(model))
println("total unserved energy = ", sum(value.(U)))

println("installed capacity (MW)")
for p in plants
    println(p, ": ", value(C[p]))
end

```

```

total system cost = 7.853529589434898e8
total unserved energy = 332.2907137925929
installed capacity (MW)
Geothermal: 285.568135230671
Coal: 0.0
NG CCGT: 1509.1728698942966
NG CT: 703.0618976373753
Wind: 2548.887923366492
Solar: 2103.7493968268923

```

3.2

We see an annual system cost of $\sim 7.58 \times 10^8$ dollars, with around 330 MWh of unserved energy

geothermal (285 MW), NG CCGT (1500 MW), NG CT (700 MW), Wind (2550 MW), solar (2100 MW), Coal (0 MW)

solution basically prioritizes both renewable and natural gas technologies because of lower emissions and competitive costs. Coal was essentially excluded under the CO2 cap. Wind and solar are supplying most of the total energy, and so it's assumed that combustion turbines would cover peak hours.

#3.3

```

genTotal = Dict()
for p in plants
    genTotal[p] = sum(value.(G[p,t]) for t in 1:numHours)
end

```



```

end

totalGen = sum(values(genTotal))
totalCap = sum(value.(C))

for p in plants
    genShare = genTotal[p] / totalGen * 100
    capShare = value(C[p]) / totalCap * 100
    println(rpad(p, 20), lpad(round(genShare, digits=2), 10), lpad(round(capShare, digits=2)
end

```

Geothermal	6.7	3.99
Coal	0.0	0.0
NG CCGT	20.3	21.11
NG CT	0.79	9.83
Wind	39.42	35.65
Solar	32.79	29.42

3.3

Wind and solar provide more than 70% of total generation, which is good because it confirms that renewables dominate under the CO2 cap. The capacity shares are high because of the variable output and lower CFs. NG CCGT contributes ~20% of total generation, which shows that it's in a middle merit source (a good in between). NG CT plants hold ~10% of installed capacity but operate rarely because they are probably used as peaking units. Finally, geothermal provides a small but pretty consistent baseload contribution. Coal is excluded due to high CO2 emissions.

Overall, the results show a system where renewables and a little bit of gas generation jointly satisfy the demand while meeting the emissions limits at minimum cost.

```

#3.5
#shadow price
dualVal = abs(dual(co2_limit))

println("Shadow price of CO2 constraint: \$", round(dualVal, digits=2), " per tonne of CO2")

#multiply by 1000 tonnes
valThousand = dualVal * 1000.0
println("Value of allowing 1000 more tonnes of CO2 emissions per year = \$", round(valThousand

```

Shadow price of CO2 constraint: \$182.77 per tonne of CO2

Value of allowing 1000 more tonnes of CO2 emissions per year = \$182771.94

3.5

The dual values of the CO2 cap is about \$182 per tCO2. I had to put it in the absolute value function because the Jump <= function.

If we allowed an additional 1000t of CO2 per year, it would reduce total cost by roughly 182000 dollars per year.

The show prices basically shows the marginal implicit carbon price – it also shows or quantifies the trade-off between cleaner generation and total economic cost in this energy optimization problem.

References

List any external references consulted, including classmates.

Emma Pereira

The built-in code fillers in VSCode also helped with syntax (mainly rounding and formatting)